

# Conservation of energy, power and efficiency

Energy  
Paper 1 · Combined and Separate  
22 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 7 starred (★) questions first.

|        | QUESTION                                                                                                                         | ANSWER                                                              | ✓                        |
|--------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|--------------------------|
| 1<br>★ | Define power.                                                                                                                    | The rate at which energy is transferred (or work is done).          | <input type="checkbox"/> |
| 2<br>★ | State the standard unit of power.                                                                                                | Watt (W)                                                            | <input type="checkbox"/> |
| 3      | State the standard unit of time.                                                                                                 | Second (s)                                                          | <input type="checkbox"/> |
| 4      | State a unit that is equivalent to 1 W.                                                                                          | 1 J/s                                                               | <input type="checkbox"/> |
| 5      | State the term for energy transferred to stores that are not useful.                                                             | Wasted energy                                                       | <input type="checkbox"/> |
| 6<br>★ | State how unwanted energy transfers can be reduced.                                                                              | By lubrication and by thermal insulation.                           | <input type="checkbox"/> |
| 7      | Lubricating moving parts reduces unwanted energy transfers to which energy store?                                                | The thermal energy store                                            | <input type="checkbox"/> |
| 8<br>★ | Describe what it means for a device to be efficient.                                                                             | Most of the input energy is transferred usefully; little is wasted. | <input type="checkbox"/> |
| 9      | State the unit of efficiency.                                                                                                    | It has no unit (it is a ratio).                                     | <input type="checkbox"/> |
| 10     | State how the efficiency of a mechanical energy transfer can be increased.                                                       | By lubricating the moving parts.                                    | <input type="checkbox"/> |
| 11     | State how the efficiency of a heating energy transfer can be increased.                                                          | By adding thermal insulation.                                       | <input type="checkbox"/> |
| 12     | Torches A and B each have an input power of 50 W. A has a useful output of 40 W and B of 45 W. Which is more efficient, and why? | Torch B.<br>Its efficiency is higher: 90%, against 80% for A.       | <input type="checkbox"/> |

| QUESTION |                                                                                                                                        | ANSWER                                                                   | ✓                        |
|----------|----------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|--------------------------|
| 13       | Heaters A and B each have a useful output of 2000 W. A has an input power of 2500 W and B of 2600 W. Which is more efficient, and why? | Heater A.<br>Its efficiency is higher: 80%, against 77% for B.           | <input type="checkbox"/> |
| 14       | State the equation linking power, energy transferred and time.                                                                         | power = energy transferred ÷ time<br>( $P = E \div t$ )                  | <input type="checkbox"/> |
| 15       | State the equation linking power, work done and time.                                                                                  | power = work done ÷ time<br>( $P = W \div t$ )                           | <input type="checkbox"/> |
| 16       | In words, state the equation for efficiency in terms of energy.                                                                        | efficiency = useful output energy transfer ÷ total input energy transfer | <input type="checkbox"/> |
| 17       | In words, state the equation for efficiency in terms of power.                                                                         | efficiency = useful power output ÷ total power input                     | <input type="checkbox"/> |
| 18<br>★  | State what is meant by dissipated energy.                                                                                              | Energy transferred to the surroundings and stored in less useful ways.   | <input type="checkbox"/> |
| 19       | State what eventually happens to wasted energy.                                                                                        | It is transferred to the surroundings, which warm up slightly.           | <input type="checkbox"/> |
| 20<br>★  | Two motors lift the same load through the same height. Motor A is quicker. Which is more powerful, and why?                            | Motor A.<br>It transfers the same energy in less time.                   | <input type="checkbox"/> |
| 21<br>★  | Explain why efficiency can never be greater than 1 (100%).                                                                             | Energy cannot be created, so useful output cannot exceed total input.    | <input type="checkbox"/> |
| 22       | A motor is 30% efficient. What does this mean?                                                                                         | 30% of the input energy is transferred usefully; 70% is wasted.          | <input type="checkbox"/> |